

That begs the question on what's to protect a nanobot assailed by particles glomming onto its surface and clogging up its gears.

The watery nanoscale environment of cell biology seems so hostile to engineering that the fact that biology works at all is almost hard to believe. But biology does work--and very well at that. The lack of rigidity, excessive stickiness, and constant random motion may seem like huge obstacles to be worked around, but biology is aided by its own design principles, which have evolved over billions of years to exploit those characteristics. That brutal combination of strong surface forces and random Brownian motion in fact propels the self-assembly of sophisticated structures, such as the sculpting of intricately folded protein molecules.

In the end, rather than ratifying the "hard" nanomachine paradigm, cellular biology casts doubt on it. But even if that mechanical-engineering approach were to work in the body, there are several issues that, in Jones' view, have been seriously underestimated by their proponents;

1. The chemical properties of the cogs and gears. They are essentially molecular clusters with odd and special shapes, but it's far from clear that they represent stable arrangements of atoms that won't rearrange themselves spontaneously.
2. A second problem has to do with the power of surface forces and the high surface area anticipated for these nanobots. Researchers attempting to shrink existing microelectromechanical systems to the nanoscale have already discovered that the combination of friction and persistent sticking can be devastating.
3. We also don't know how an intricate arrangement of cogs and gears that depends on precision and rigidity to work will respond to thermal noise and Brownian bombardment at room temperature

All these complications when put together suggests that the range of environments in which rigid nanomachines could operate, if they operate at all, would be quite limited. If, for example, such devices can function only at low temperatures and in a vacuum, their impact and economic importance would be virtually nil.

Jones advocates more attention to "soft" nanotech, which is nanomachinery with similar design principles to biology -- the biomimetic approach -- as the most plausible means of making progress in nanotech. He asserts that human will reap major medical advances by radically reengineering existing microorganisms, especially in nanodevices that perform integrated diagnosis and treatment of some disorders

Intrinsic top-down unmanufacturability

The last paper we will be taking up was authored by Professor Mike Kelly-Centre for Advanced Photonics and Electronics, University of Cambridge -- one of a leading nanotechnology scientist. This paper entitled "Intrinsic top-down